

HotelRevAI

AI-Driven Revenue Analysis for Hotels

Final Internship Project Report

Student: Rinal Dave

Program: Infosys Springboard Virtual Internship

Project Type: Data Analysis & Visualization

Tools Used: Microsoft Power BI, Power Query, DAX

Mentor: Selciya

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1. Abstract

The **HotelRevAI project** focuses on analyzing hotel booking data and generating actionable insights using business intelligence tools. Using **Microsoft Power BI**, the dataset was cleaned, modeled, and analyzed to create interactive dashboards that support hotel revenue analysis.

The project transforms raw booking data into meaningful insights related to **booking patterns, customer segmentation, revenue performance, and seasonal demand**. These insights help hotel management make **data-driven decisions** for improving operational efficiency and maximizing revenue.

2. Introduction

The hospitality industry generates a large amount of booking and customer data every day. Analyzing this data effectively helps hotel management make better **operational and financial decisions**.

The **HotelRevAI – AI Driven Revenue Analysis for Hotels** project focuses on analyzing hotel booking data and converting it into meaningful business insights using **data analytics and visualization techniques**.

The system uses **Microsoft Power BI along with DAX calculations and AI capabilities** to transform raw booking data into interactive dashboards that support revenue analysis, booking behavior understanding, and performance monitoring.

The final outcome is a set of dashboards that provide insights into:

- Booking trends
- Revenue distribution
- Customer behavior
- Operational performance
- Seasonal demand patterns

These insights help hotel managers **optimize pricing strategies, improve occupancy rates, and enhance customer experience**.

3. Project Objectives

The main objectives of the project are:

- Analyze hotel booking and revenue data
- Identify booking patterns and seasonal trends
- Create key performance indicators (KPIs)
- Develop interactive dashboards using Power BI
- Apply DAX for business metric calculations
- Provide actionable insights for hotel management

4. Dataset Description

The dataset used in this project was obtained from **Kaggle** and contains hotel booking records from **2018–2020**.

The dataset includes:

- Customer information
- Reservation details
- Booking channels
- Market segments
- Stay duration
- Revenue attributes

This data enables analysis of **customer behavior, revenue performance, booking patterns, and operational trends**.

5. Data Dictionary

Important fields used in the dataset include:

- **hotel** – Name/type of the hotel where booking was made
- **is_canceled** – Indicates if the booking was canceled (1) or not (0)
- **lead_time** – Number of days between booking and arrival date
- **arrival_date_year** – Year of arrival

- **arrival_date_month** – Month of arrival
- **stays_in_weekend_nights** – Number of weekend nights stayed
- **stays_in_week_nights** – Number of weekday nights stayed
- **adults** – Number of adults in the booking
- **children** – Number of children in the booking
- **meal** – Type of meal plan selected
- **country** – Guest country of origin
- **market_segment** – Booking market segment
- **distribution_channel** – Booking channel used
- **customer_type** – Type of customer (Transient, Contract, Group etc.)

6. Methodology and Project Workflow

The project followed several structured stages:

1. Data Collection

The hotel booking dataset was collected from Kaggle and imported into Power BI.

2. Data Cleaning

Raw data was cleaned to remove inconsistencies and ensure accuracy.

3. Feature Engineering

New calculated columns were created to support deeper analysis.

4. Data Modeling

The dataset was structured using a **Star Schema** to support efficient reporting.

5. DAX Calculations

Business metrics and KPIs were created using **Data Analysis Expressions (DAX)**.

6. Dashboard Development

Interactive dashboards were developed in Power BI.

7. AI-Driven Analysis

Analytics techniques were used to identify patterns and trends in booking data.

7. Data Cleaning and Feature Engineering

Data preprocessing involved:

- Removing duplicate records
- Handling missing values
- Correcting data types
- Formatting date columns

Calculated Columns

Total Nights

Total Nights = stays_in_week_nights + stays_in_weekend_nights

Total Revenue

Total Revenue = ADR × Total Nights

Date Features

Additional date attributes were extracted:

- Year
- Month
- Day

These fields help analyze **time-based booking trends**.

Cancellation Indicator

Bookings where **is_canceled = 0** were considered successful bookings.

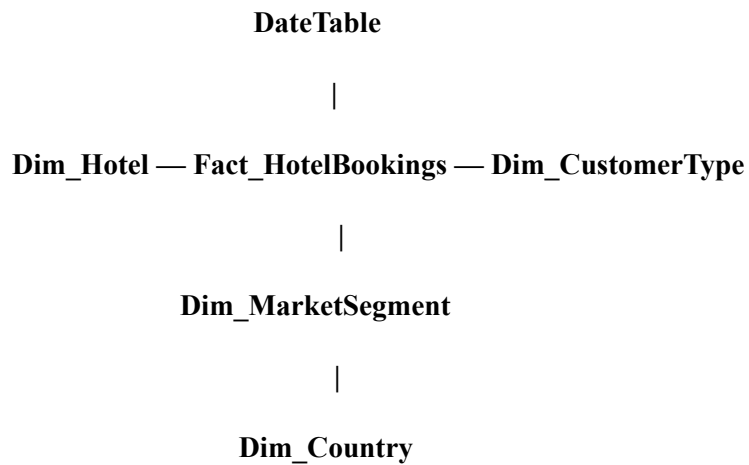
8. Data Modeling Architecture

The project uses a **Star Schema Data Model**, which is widely used in business intelligence systems.

The model separates data into:

- **Fact Tables** – containing transactional data
- **Dimension Tables** – containing descriptive attributes

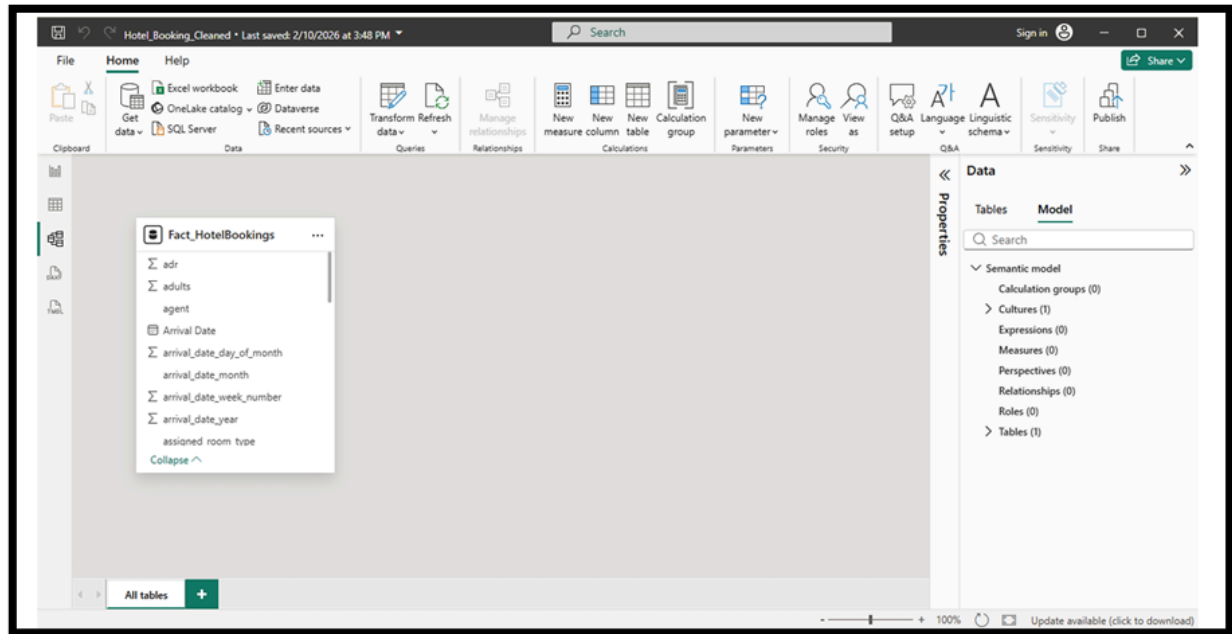
Fact Table



8.1 : Fact_HotelBookings

Contains transactional booking data such as:

- Arrival Date
- Hotel
- Country
- Customer Type
- Market Segment
- Booking metrics
- Revenue values



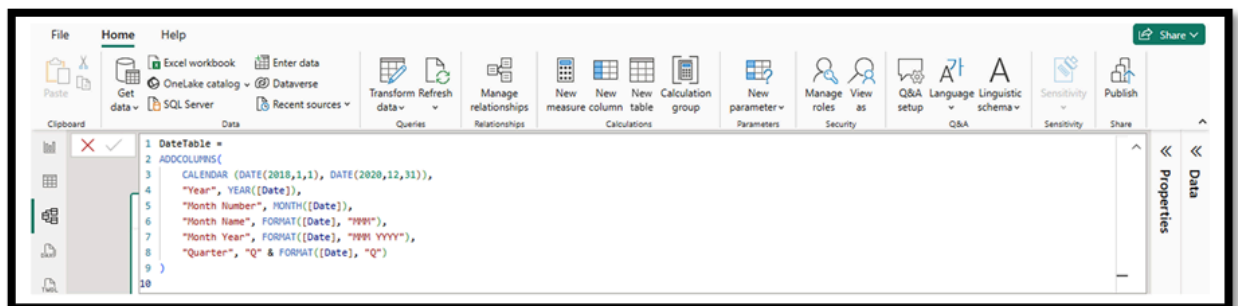
8.2 : Dimension Tables

Several dimension tables were created to organize descriptive data.

8.3 : DateTable

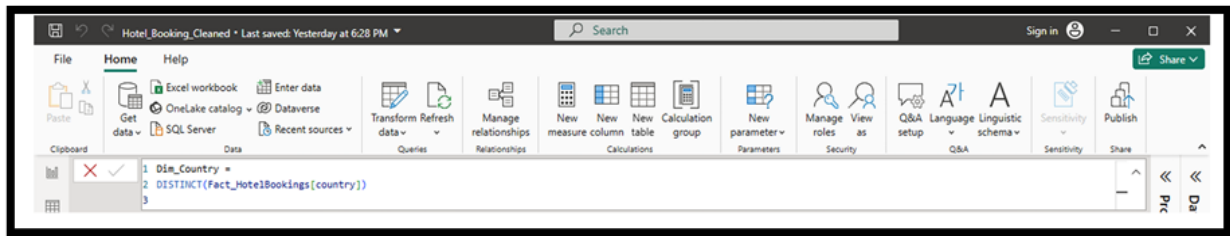
Used for time-based analysis.

- Date
- Month
- Month Number
- Quarter
- Year



8.4 : Dim_Country

- Contains unique country names for geographic analysis.
- Used to analyze bookings by geographic location.



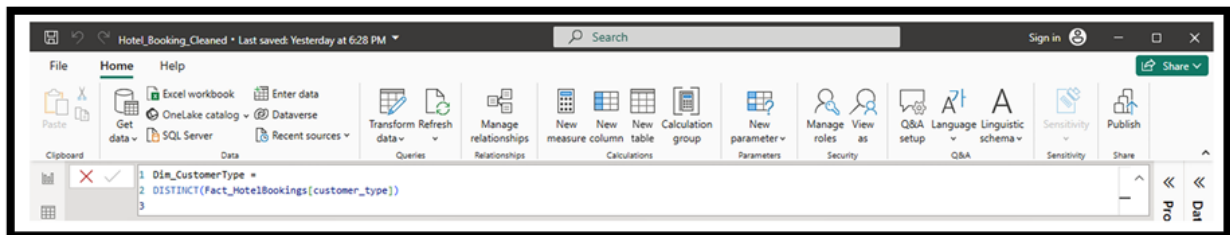
8.5 : Dim_CustomerType

Stores different categories of customers.

Examples:

- Transient
- Contract
- Group

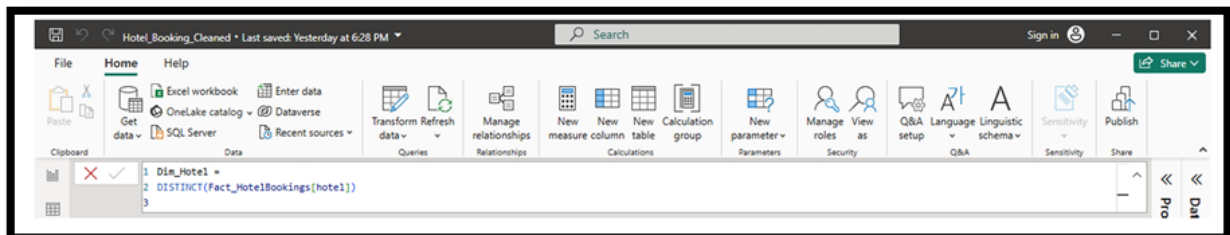
This helps analyze booking behavior across customer segments.



8.6 : Dim_Hotel

Stores hotel names for performance comparison

Used for hotel-level performance comparison.



8.7 : Dim_MarketSegment

Examples include:

- Direct

- Corporate
- Online Travel Agency
- Offline Travel Agent

This helps analyze booking channels.

9. ER Diagram

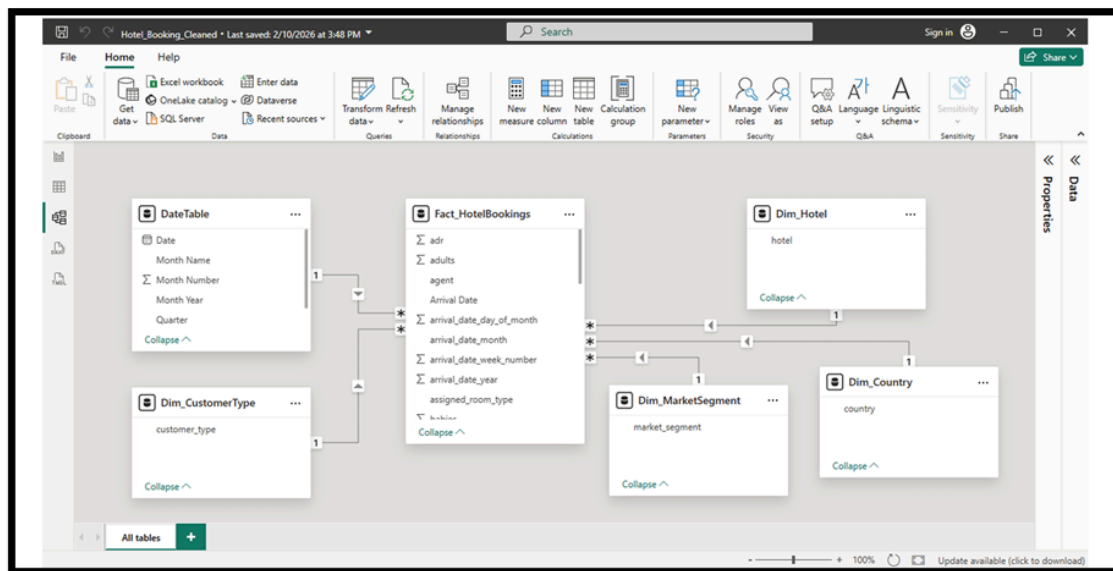
The ER Diagram represents relationships between fact and dimension tables.

Relationships:

- DataTable[Date] → Fact_HotelBookings[Arrival Date]
- Dim_Country[Country] → Fact_HotelBookings[Country]
- Dim_CustomerType[Customer Type] → Fact_HotelBookings[Customer Type]
- Dim_Hotel[Hotel] → Fact_HotelBookings[Hotel]
- Dim_MarketSegment[Market Segment] → Fact_HotelBookings[Market Segment]

All relationships are:

- Active
- One-to-Many (1 → *)
- Single-direction filtering



10. DAX Calculations

Total Bookings

Total Bookings = COUNT(Bookings[Booking_ID])

Total Revenue

Total Revenue = SUM(Bookings[Revenue])

Weekend Revenue

Weekend Revenue =
CALCULATE(SUM(Bookings[Revenue]),Bookings[Day_Type] = "Weekend")

Weekday Revenue

Weekday Revenue =
CALCULATE(SUM(Bookings[Revenue]),Bookings[Day_Type] = "Weekday")

Average Revenue per Booking

Average Revenue per Booking =
DIVIDE([Total Revenue], [Total Bookings], 0)

Cancellation Rate

Cancellation Rate =
DIVIDE(CALCULATE(COUNTROWS(HotelData), HotelData[is_canceled] = 1),COUNTROWS(HotelData))

Revenue per Available Room (RevPAR)

RevPAR =
DIVIDE(SUM(Bookings[Revenue]),COUNT(Bookings[Booking_ID]),0)

Average Length of Stay

Average Length of Stay = AVERAGE(Bookings[Total Nights])

Total Stay Nights

Total Stay Nights =
SUM(Bookings[stays_in_week_nights]) + SUM(Bookings[stays_in_weekend_nights])

Average Guests per Booking

Average Guests per Booking =
AVERAGE(Bookings[adults] + Bookings[children])

11. Business Metrics and Key Insights

The analysis of the hotel booking dataset generated several important business metrics that help evaluate revenue performance, booking patterns, and customer behavior.

Key Business Metrics

- **Total Bookings:** Measures the total number of reservations made during the analysis period.
- **Total Revenue:** Represents the total income generated from all hotel bookings.
- **Average Daily Rate (ADR):** Indicates the average revenue earned per room per night.
- **Average Revenue per Booking:** Shows the average amount of revenue generated from each reservation.
- **Revenue per Available Room (RevPAR):** Combines pricing and occupancy performance to measure overall hotel revenue efficiency.
- **Weekend Revenue:** Measures revenue generated from weekend stays.
- **Weekday Revenue:** Measures revenue generated from weekday stays.
- **Cancellation Rate:** Indicates the percentage of bookings that were cancelled.
- **Average Length of Stay:** Represents the average number of nights guests stay in the hotel.
- **Average Lead Time:** Measures how early customers make reservations before their arrival date.
- **Repeat Guest Rate:** Shows the proportion of returning customers.
- **Total Stay Nights:** Represents the total number of nights booked by guests.

Key Insights from the Analysis

- **Online Travel Agencies (OTA)** contribute the largest share of bookings and revenue.
- **City Hotels generate more revenue than Resort Hotels**, indicating higher demand in urban locations.

- The **average length of stay is relatively short**, suggesting that most guests stay for only a few days.
- **Weekday revenue is higher than weekend revenue**, indicating strong demand from business travelers.
- **Cancellation rate is significant**, which may impact revenue stability.
- **Most customers are new guests**, while repeat guests represent a smaller proportion.
- **Seasonal trends exist**, with certain months showing higher booking volumes.
- **Lead time analysis shows many bookings are made close to the arrival date**, indicating short planning cycles for travelers.
- **Guests from multiple countries contribute to hotel bookings**, highlighting an international customer base.

12. Dashboard Development

Four analytical dashboards were developed in **Power BI**.

1. Executive Overview Dashboard

Purpose

Provides a **high-level summary of hotel performance** from 2018 to 2020 and helps management quickly understand overall business performance.

Key Metrics

- **Net Revenue:** 30.5M
- **Average ADR (Average Daily Rate):** 100.8
- **Total Bookings:** 86.8K
- **Average Days of Stay:** 3.5 days
- **Guests in Hotels:** 168.7K

Visualizations and Insights

1. Total Revenue by Year

- Shows yearly revenue performance.

- Helps identify growth or decline in revenue between 2018–2020.

2. Total Revenue by Hotel

- Compares revenue generated by **City Hotel** and **Resort Hotel**.
- Helps determine which hotel type performs better financially.

3. Total Bookings by Year

- Displays the number of bookings per year.
- Helps analyze customer demand trends.

4. Revenue by Market Segment

- Shows revenue contribution from different segments:
 - Online TA or Offline TA/TO
 - Direct
 - Groups
 - Aviation
 - Complementary
- Helps identify the most profitable booking channel.

Filters

- Arrival Date
- Hotel Type
- Market Segment



2. Booking Behavior Analysis Dashboard

Purpose

Analyzes **customer booking patterns and guest behavior** to understand how customers interact with hotel services.

Key Metrics

- Net Revenue
- Average ADR
- Total Bookings
- Average Stay
- Total Guests

Visualizations and Insights

1. Lead Time by Quarters

- Shows how early customers make reservations.
- Helps understand booking trends throughout the year.

2. New vs Repeated Guests

- Compares new customers and returning customers.
- Data shows most guests are **new customers (96.59%)**.

3. Customer Type Distribution

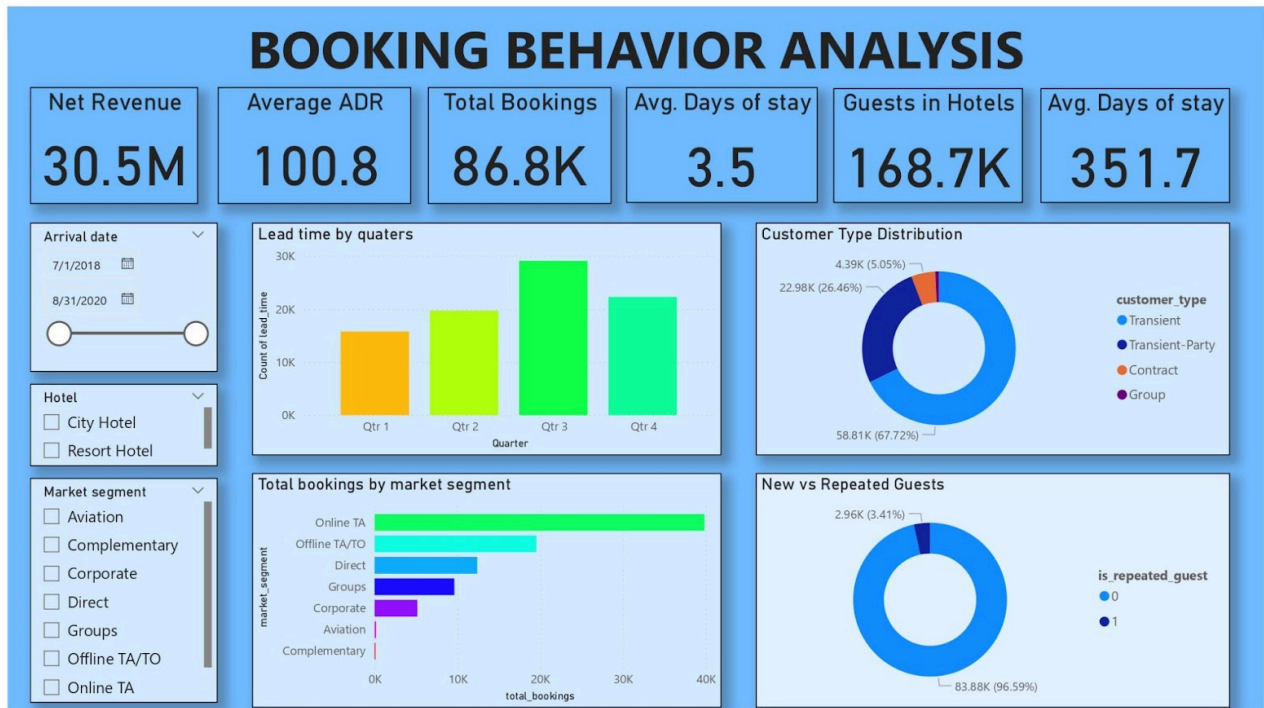
- Displays booking distribution by customer types:
 - Transient
 - Transient-Party
 - Contract
 - Group
- Helps understand which customer type books the most.

4. Total Bookings by Market Segment

- Shows which booking channels bring the most reservations.

Filters

- Arrival Date
- Hotel Type
- Market Segment



3. Revenue & Pricing Analysis Dashboard

Purpose

Focuses on **revenue generation and pricing strategy analysis** to help improve profitability.

Key Metrics

- Net Revenue
- Average ADR
- Total Bookings
- Average Days of Stay
- Guests in Hotels

Visualizations and Insights

1. Total Revenue by Quarter

- Displays revenue distribution across quarters.
- Helps identify seasonal revenue patterns.

2. Total Revenue by Hotel

- Compares revenue between City Hotel and Resort Hotel.
- Helps determine which hotel generates higher income.

3. Revenue Trend Over Time

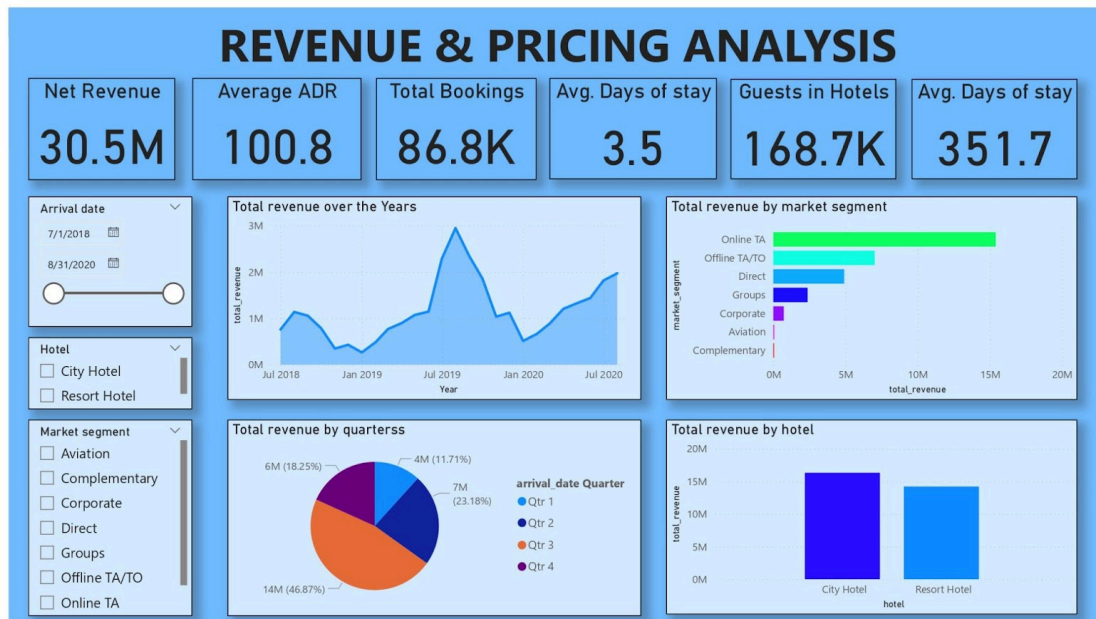
- Shows revenue changes over months and years.
- Useful for tracking business growth.

4. Revenue by Market Segment

- Identifies which booking channels contribute most revenue.

Filters

- Arrival Date
- Hotel Type
- Market Segment



4. Operational Insights Dashboard

Purpose

Provides insights into **hotel operations**, **booking patterns**, and **customer service trends**.

Visualizations and Insights

1. Special Requests by Year

- Shows number of special requests from guests.
- Helps improve customer service planning.

2. Total Stays Over the Month

- Displays the total number of nights stayed each month.
- Helps identify seasonal occupancy trends.

3. Monthly Bookings Trend

- Shows how bookings vary month by month.
- Helps plan staff and resources.

4. Weekend vs Weekday Revenue

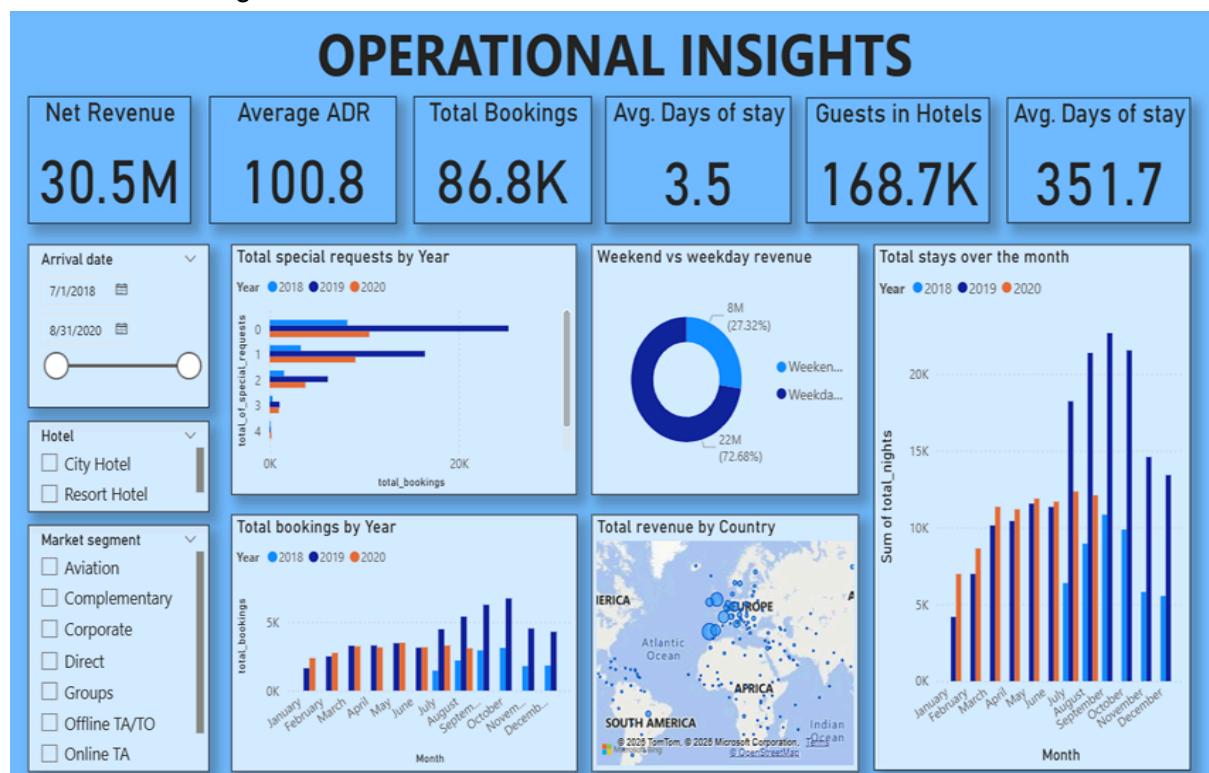
- Compares revenue generated during weekdays and weekends.
- Shows that weekday revenue is higher.

5. Revenue by Country (Map)

- Shows geographical distribution of guests.
- Helps identify key international markets.

Filters

- Arrival Date
- Hotel Type
- Market Segment



13. Tools and Technologies Used

The project used the following tools:

- Microsoft Power BI
- Power Query
- DAX
- Excel Dataset
- Data Modeling Techniques

14. Project Outcomes

The project successfully developed a **data-driven analytics system** that converts hotel booking data into interactive dashboards and actionable insights.

The dashboards help stakeholders:

- Monitor revenue performance
- Understand booking behaviour
- Analyze operational trends
- Support strategic decision making

15. Conclusion

The **HotelRevAI system** demonstrates how **data analytics and visualization** can support revenue optimization in the hospitality sector.

By transforming raw booking data into interactive dashboards, the system enables hotel managers to understand **customer behavior, revenue patterns, and operational efficiency**.

The project highlights the importance of **data-driven decision making in the hospitality industry**.

16. References

- Kaggle – Hotel Booking Dataset
- Microsoft Power BI Documentation